THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** a feather	Air resistance matters for feathers.
2	**(c)** 20 per cent	30/150.
3	**(d)** 10 g	Conservation of mass.
4	**(a)** 16	Sum of electrons in a neutral atom.
5	**(c)** slowly stops swinging	Friction at the pivot and air resistance.
6	**(a)** fractional distillation	Boiling-point difference.
7	**(d)** $Al_2(SO_4)_3$	Al3+ and SO4 2-.
8	**(a)** 12	2, 8, 2.
9	**(a)** 2500	1 g/cm3 = 1000 kg/m3.
10	**(d)** sublimation, dissolving and filtering, evaporation	Camphor first.
11	**(c)** Mg and Cl	Metal and non-metal.
12	**(c)** 35.5 u	26.25 + 9.25.
13	**(a)** 8 m/s	100/12.5.
14	**(a)** 10 g	4/100 x 250.
15	**(b)** 28 g	0.8235 x 34.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Acceptable: air hockey puck on a smooth table (1); Fails: car braking on a rough road (1).	1+1
17	8/100 x 500 (1); = 40 g (1).	1+1
18	Mass of water = 18 g (1); Reactants = 18 g, products = 18 g (1).	1+1
19	A = 24 (1); 2, 8, 2 (1).	1+1
20	Easy to read and compare (1); 3 x 10^8 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	T = 32/20 = 1.6 s (1); Reduces reaction-time error (1); Gives better precision (1).	1+1+1
22	36/200 x 100 = 18 per cent (1); New solution = 100 g (1); New percentage = 36 per cent (1).	1+1+1
23	C = 3/11 x 55 = 15 g (1); O = 8/11 x 55 = 40 g (1); Total = 55 g (1).	1+1+1
24	A = 40 (1); 2, 8, 8, 2; valency 2 (1); Calcium (1).	1+1+1
	OR: Atom about 10^-10 m (1); Sun about 1.5 x 10^11 m (1); Easy comparison (1).	1+1+1
25	KNO3 (1); Its solubility falls sharply on cooling (1); NaCl solubility changes little (1).	1+1+1
	OR: NaCl ionic, CCl_4 covalent (1); NaCl high melting, CCl_4 low (1); NaCl conducts in solution, CCl_4 does not (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Periods: 1.0, 1.42, 1.74, 2.0 s (1); T increases with length (1); Increase is not proportional (1); No friction or air resistance assumed (1); Small amplitude assumed (1).	1+1+1+1+1
	OR: A = 5 per cent (1); B = 10 per cent (1); C = 15 per cent (1); Mixing equal masses of A and C gives 10 per cent (1); Ranking C > B > A (1).	1+1+1+1+1

27	Ratio H : O in the first sample = 1 : 8 (1); Ratio in the second sample = 2 : 16 = 1 : 8 (1); The ratio is the same for different samples (1); Law: a pure compound always contains the same elements in a fixed ratio by mass (1); Proust gave this law; it distinguishes a compound from a mixture (1).	1+1+1+1+1
	OR: p = e = 13, n = 14 (1); 2, 8, 3 (1); Valency 3 (1); Al^{3+} (1); Al_2O_3 (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 1	2, 8, 8, 1.
2	**(a)** supports the law	Agreement supports it.
3	**(b)** sublimation	Iodine sublimes.
4	**(b)** $(NH_4)_3PO_4$	NH4+ and PO4 3-.
5	**(c)** 4	2, 8, 4.
6	**(a)** 20	72 x 5/18.
7	**(a)** is collected first	Evaporates first.
8	**(d)** copper	Metals have free electrons.
9	**(d)** 10.8 u	2 + 8.8.
10	**(c)** 16.7 m/s	60 km/h x 5/18.
11	**(a)** 50 mL	0.2 x 250.
12	**(a)** 30 g	20 x 60/40.
13	**(b)** calcium	Z = 20.
14	**(d)** low pressure and high temperature	Real gases approach ideal behaviour.
15	**(a)** separating funnel	Immiscible.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	To measure the temperature of the vapour (1); Which equals the boiling point of the liquid (1).	1+1
17	H has valency 1 and O has valency 2 (1); Two H atoms are needed for one O atom: H_2O (1).	1+1
18	(63 x 70 + 65 x 30)/100 (1); = 63.6 u (1).	1+1
19	Hypothesis: proposed explanation (1); Prediction: expected result from the hypothesis (1).	1+1
20	Dissolved salts raise the boiling point (1); Solute particles reduce vapour pressure (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	2, 8, 3 (1); Z = 13 (1); Valency 3 (1).	1+1+1
22	Points follow a trend (1); A stray point stands out (1); Check and repeat (1).	1+1+1
23	Pure liquid has a fixed boiling point (1); Mixture has components with different b.p. (1); Composition changes (1).	1+1+1
24	O is 30 per cent = 48 u (1); 2X = 70/30 x 48 = 112 (1); X = 56 u (iron) (1).	1+1+1
	OR: Electrons jump between shells (1); Photons of fixed energy are emitted or absorbed (1); Gives lines in the spectra (1).	1+1+1
25	0.07 mm = 7 x 10^-5 m (1); = 70 micrometres (1); 1 mm = 1000 micrometres (1).	1+1+1
	OR: Acetic acid = 25 mL (1); Add to some water (1); Make up to 250 mL (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Initial observation (1); Hypothesis (1); Repeated experiments (1); Peer review (1); Wide acceptance (1).	1+1+1+1+1
	OR: Air filtered, CO2 and water removed (1); Compressed and cooled (1); Liquid air fractionated (1); Nitrogen, argon, oxygen collected (1); Uses of each gas (1).	1+1+1+1+1
27	O = 1.5 g (1); Cu : O = 6 : 1.5 = 4 : 1 (1); CuO ratio = 63.5 : 16 = 3.97 : 1 (1); Agrees within error (1); Compound confirmed as CuO (1).	1+1+1+1+1

	OR: C-12: 6 p, 6 n (1); C-13: 7 n (1); C-14: 8 n, radioactive (1); Dating of fossils by decay of C-14 (1); Half life about 5730 years (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** $Ca(OH)_2$	Ca2+ and OH-.
2	**(d)** aluminium	2, 8, 3.
3	**(d)** 0.15	0.5 x 0.3.
4	**(b)** colloidal particles	Particles smaller than pores.
5	**(b)** O_2	Two shared pairs.
6	**(a)** neutrons	Mass number differs.
7	**(d)** 0.2	2 x 0.5 x 0.2.
8	**(c)** 470 g	Solute 30 g.
9	**(d)** 12 g	9 x 4/3.
10	**(a)** 2, 8	Loses 2.
11	**(b)** defines its limits and develops a better model	Model improvement.
12	**(d)** crystallisation	Cool saturated solution.
13	**(b)** Fe_2O_3	Fe3+ and O2-.
14	**(c)** carbon (2, 4)	4 for C; others 2, 1, 1.
15	**(d)** 12.6 g	Average.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	12/44 x 100 (1); = 27.3 per cent (1).	1+1
17	Z = 24 - 12 = 12 (1); 2, 8, 2 (1).	1+1
18	Sum = 25.5 (1); Mean = 5.1 g (1).	1+1
19	Solution = 100 g (1); 12 per cent (1).	1+1
20	H : O = 1 : 8 (1); H = 4 g, O = 32 g (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	N = 6.6 - 4.8 = 1.8 g (1); Ratio = 4.8 : 1.8 (1); = 8 : 3 (1).	1+1+1
22	Both: p = e = 17 (1); n = 18 and 20 (1); Isotopes (1).	1+1+1
23	Distance light travels in one year (1); About 9.46 x 10^15 m (1); Used for stars because they are very far (1).	1+1+1
24	Size (1); Tyndall effect (1); Examples: sugar solution, milk (1).	1+1+1
	OR: Ions are fixed in the solid (1); On melting the ions become free (1); Free ions carry charge (1).	1+1+1
25	Iodine-131 is absorbed by the thyroid, treats goitre (1); Cobalt-60 gives gamma rays (1); Used in cancer therapy (1).	1+1+1
	OR: Wear goggles: protect the eyes (1); Do not taste: poisons (1); Add acid to water slowly: avoid splashes (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Air breathed in a day (1); Heartbeats in a life (1); Mass of water in a school tank (1); Assumed average values (1); Uncertainty from approximations (1).	1+1+1+1+1
	OR: Magnet: iron (1); Sublimation: ammonium chloride (1); Dissolve and filter: sand (1); Evaporate: salt (1); Principles: magnetic, sublimation, solubility, boiling point (1).	1+1+1+1+1
27	Na: 2, 8, 1 loses 1 (1); Mg loses 2, O gains 2: MgO (1); Ca loses 2; each F gains 1: CaF2 (1); Noble gas arrangement (1); Formulae by balance of charges (1).	1+1+1+1+1

	OR: Let x per cent be Cl-35 (1); 35x + 37(100 - x) = 3550 (1); x = 75 (1); Cl-35 75 per cent, Cl-37 25 per cent (1); Mixture of isotopes gives non-integer mass (1).	1+1+1+1+1

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